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Prognostic score matching methods for estimating the average effect of a non-reversible binary time-dependent treatment on the survival function

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Abstract

In evaluating the benefit of a treatment on survival, it is often of interest to compare post-treatment survival with the survival function that would have been observed in the absence of treatment. In many practical settings, treatment is time-dependent in the sense that subjects typically begin follow-up untreated, with some going on to receive treatment at some later time point. In observational studies, treatment is not assigned at random and, therefore, may depend on various patient characteristics. We have developed semi-parametric matching methods to estimate the average treatment effect on the treated (ATT) with respect to survival probability and restricted mean survival time. Matching is based on a prognostic score which reflects each patient's death hazard in the absence of treatment. Specifically, each treated patient is matched with multiple as-yet-untreated patients with similar prognostic scores. The matched sets do not need to be of equal size, since each matched control is weighted in order to preserve risk score balancing across treated and untreated groups. After matching, we estimate the ATT non-parametrically by contrasting pre- and post-treatment weighted Nelson–Aalen survival curves. A closed-form variance is proposed and shown to work well in simulation studies. The proposed methods are applied to national organ transplant registry data.

Keywords Landmark analysis · Causal inference · Matching · Prognostic score · Semiparametric method · Survival function · Treatment effect

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1 Introduction

Treatments are often evaluated in terms of survival time. In many practical settings, the treatment is time dependent, such that patients begin follow up untreated, with some patients subsequently receiving treatment and others dying first. We consider the setting wherein treatment can be cast as a non-reversible binary (0/1) indicator function. This set-up has a long history in survival analysis, dating back to the seminal analysis of the Stanford heart transplant data (Cox and Oakes 1984). A diagram of this set-up is given in Fig. 1, where ($t = 0$) represents the start of follow up, T denotes treatment time and D denotes death time. The diagram pertains to the uncensored data, as evidenced by the fact that right censoring time C is not represented.

The solid organ transplantation setting aligns with the data structure described in the preceding paragraph. Patients begin follow-up at the time of wait listing and are untreated (i.e., not transplanted), with some going on to receive a transplant during follow up. Our motivating example is analogous. We consider deceased-donor liver transplant (LT) recipients, who are at increased risk for developing kidney issues, including end-stage renal disease (ESRD). In our case, the study population includes LT recipients who are followed from the time of LT forward. Some of these LT recipients will have some form of kidney dysfunction at the time of LT, while others will develop such issues after LT. As such, a small but not negligible number of LT recipients will subsequently receive a kidney transplant (KT); this has been referred to as kidney-after-liver transplantation (KALT). In addition to the obvious clinical implications, this practice has important health policy implications. In particular, given the high demand for donor kidneys among ESRD patients, one could certainly question the value of performing KT on a patient that has already been fortunate enough to receive a LT. An understandably skeptical view of the practice is fueled by a recent study by Sharma et al. (2016) which demonstrated that simultaneous liver–kidney transplantation (SLKT) provided minimal survival benefit to patients, relative to LT alone. Although SLKT and KALT are fundamentally different, a null finding for SLKT with respect to survival benefit increases the importance of evaluating KALT. Few evaluations of KALT have employed requisite statistical rigor for drawing reliable conclusions. It is important at this juncture to note that we have excluded SLKT recipients, such that the KT considered in our analysis necessarily occurs after time 0 (i.e., with $t = 0$ corresponding to the time of LT, for our purposes). A state diagram corresponding to our motivating example is shown in Fig. 2, with the data structure paralleling that from Fig. 1.

Cox regression (Cox 1972) is frequently employed for survival analysis in the presence of covariate adjustment. The key parameter in such an analysis is the hazard ratio (HR). Despite the utility of such an approach (as evidenced abundantly by its popularity), the approach has several limitations. In the absence of proportional hazards, the HR is a rather artificial parameter, with its limiting value only defined implicitly and depending on the censoring distribution (Struthers and Kalbfleisch 1986). Time-dependent versions of the HR, from the so-called Cox nonproportional hazards model (Kalbfleisch and Prentice 2002), are difficult to interpret, as is now well documented (e.g., see Wei and Schaubel 2008). Moreover, even if proportionality holds, most clinical investigators would prefer to compare survival curves than hazard functions.

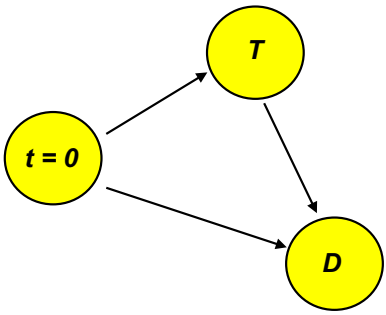

Fig. 1 State diagram: ($t = 0$) refers to the start of follow-up; T denotes treatment and D denotes death

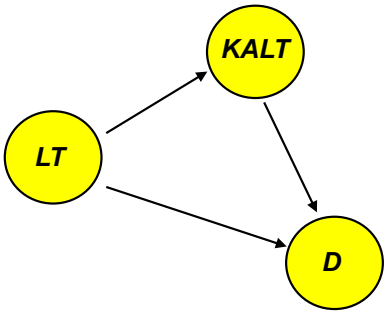

Fig. 2 State diagram (motivating example): time of liver transplantation (LT) represents time 0, with the treatment being kidney-after-liver transplantation (KALT)

The majority of methods for analyzing the effect of time-dependent (i.e., binary, non-reversible) treatments on survival employ a time-dependent treatment indicator (Robins et al. 2000; Hernán et al. 2000; Petersen et al. 2007). This is largely due to the fact that there are a few existing methods to compare survival functions when treatment is time-dependent.

In this report, we propose methods to evaluate a time-dependent treatment with respect to the survival function. Specifically, we estimate the average effect of treatment on the treated (ATT). The proposed methods involve matching each treated patient to a set of similar yet-untreated patients, with similarity defined in terms of a prognostic score (Hansen 2008). A variant of Inverse Probability of Censoring Weighting (IPCW; Robins and Rotnitzky 1992) is utilized, in order to accomplish two goals: (i) account for the fact that treatment times themselves are subject to censoring, and (ii) accommodate the fact that matched yet-untreated subjects may subsequently receive treatment. The proposed method is an extension of Li et al. (2014), with the major advances being (a) matching multiple matches per treated patient, as opposed to 1:1 matching (b) the estimation of restricted mean survival time (area under the survival curve). The use of $1 : m$ matching, with $m > 1$, requires rather delicate modification of the weight function, as we detail in Sect. 2. We demonstrate the great efficiency gains for the proposed methods, relative to 1:1 matching, through simulations and in the real-data application.

There are now several existing methods to evaluate the causal effect of a time-dependent treatment in the context of a censored survival outcome; e.g., see the summary by Robins and Hernán (2008). Unlike most existing works, our proposed methods specifically target the ATT, and define the ATT through the difference in

(and area between) the survival functions. Unlike the average causal effect (ACE), the ATT is particularly well-suited to the study of organ transplantation since it would be impossible and undesirable to transplant all patients. Expressing the treatment effect in terms of survival functions would be appealing to clinical investigators. If treatment was randomly assigned at time 0, treatment effects are often illustrated through a plot of the treated versus placebo Kaplan and Meier (1958) curves. Our methods are intended to yield an analogous plot constructed through techniques that accommodate the use of observational data and in the context of a treatment assigned after time 0 and in the absence of randomization.

With respect to existing methods, marginal structural models (Robins et al. 2000; Hernán et al. 2000, 2001; Petersen et al. 2007) typically estimate the ACE through the causal hazard ratio (HR). Structural nested failure time models (SNFTMs) often use the accelerated failure time model as the basis for a time-dependent treatment effect defined as a ratio of mean survival times; e.g., Robins and Rotnitzky (1992), Lok et al. (2004) and Hernán et al. (2005). As an ACE measure, the hazard ratio is problematic due to non-collapsibility. Each of the above listed methods fully parametrizes the treatment effect, with associated advantages including conciseness (e.g., one-number summary of ACE), at the expense of inaccuracy due to model mis-specification. Although the methods we propose involve regression modeling, the treatment effect is estimated non-parametrically. Our current proposal is not intended to replace any of the afore-cited works, but to propose an alternative set of procedures which should be easy for practitioners to implement, and which yields results that are straightforward to interpret.

The remainder of this report is organized as follows. In Sect. 2, the afore-described data structure is formalized, and the proposed methods are described. Asymptotic properties are provided in Sect. 3. A simulation study is presented in Sect. 4. The proposed methods are applied to estimate the survival benefit of KALT among LT recipients in Sect. 5. The report concludes with a discussion in Sect. 6.

2 Proposed methods

We begin by setting up the requisite notation. We then formalize the treatment effect of interest as an average difference in survival with respect to counterfactual variates. The proposed matching techniques are then described, followed by a description of the ATT estimators.

2.1 Notation and ATT measure

Fist, we set up notation for the observed data (with that for the counterfactuals to occur later). Let D_i denote the death time and C_i the censoring time for subject i, for $i = 1, \ldots, n$. The observation time is denoted by $X_i = D_i \wedge C_i$, where $a \wedge b = \min\{a, b\}$. The death indicator is then given by $\Delta_i = I(D_i \leq C_i)$ where $I(A)$ is an indicator function taking the value 1 when event A is true and 0 otherwise. The at-risk indicator is defined as $Y_i(t) = I(X_i \geq t)$. Let $\boldsymbol{Z}_i$ be the covariate vector, which is assumed

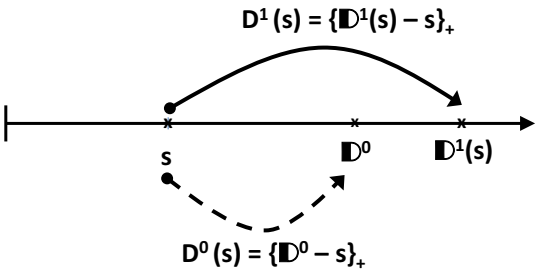

Fig. 3 Counterfactuals and time scale, as defined in Sect. 2.1

to not depend on time. The treatment time is represented by T_i, with corresponding indicator $\Delta_i^T = I(T_i \leq X_i)$. The observed data consist of $(X_i, T_i, \Delta_i, \Delta_i^T, \boldsymbol{Z}_i)$ for $i = 1, \ldots, n$, which are assumed to be independent and identically distributed. We assume that, given $\boldsymbol{Z}_i$, D_i and T_i are independently censored by C_i.

We now set up the notation for the counterfactual variates, through which the treatment effect will be defined. We temporarily suppress the subscript i, with the variates then referencing any hypothetical subject. Consider the setting where a binary non-reversible treatment is applied at follow-up time s, if assigned to a patient. Let $\mathbb{D}^0$ represent the counterfactual death time if treatment is not (meaning never) assigned to the patient. We let $\mathbb{D}^1(s)$ denote the counterfactual death time if treatment is assigned. Note that the time scale underlying $\mathbb{D}^1(s)$ is total time (i.e., since follow-up began). We then define the corresponding counterfactual post-treatment residual survival times, $D^1(s) = \{\mathbb{D}^1(s) - s\}_+$ and $D^0(s) = \{\mathbb{D}^0 - s\}_+$, where $a_+ = \max\{a, 0\}$. Here, $D^1(s)$ represents residual time survived after treatment (i.e., measured from the time of treatment initiation). The counterfactual $D^0(s)$ represents the residual survival time after time s in the absence of treatment. Naturally, only patients alive at follow-up time s can be treated at that time. The notation described in this paragraph is depicted in Fig. 3.

Replacing s in the preceding paragraph with T, consider a patient, who initiates treatment at time T. Let $D^1(T)$ denote the potential residual survival time following treatment assignment; i.e., the post-treatment survival time, $D^1(T) = \{\mathbb{D}^1(T) - T\}_+$, Conversely, let $D^0(T)$ denote the potential remaining survival time had the patient not received treatment; i.e., survival in the absence of treatment, $D^0(T) = \{\mathbb{D}^0 - T\}_+$. For a treated subject, $D^0(T)$ represents the residual time survived (after time T) if in fact treatment had not been assigned. Naturally, although $D^0(T)$ can never be observed, it is an essential variate to consider in the context of estimating the causal effect of treatment. Our analysis will be based on an appropriate average across the distribution of $D^1(T) - D^0(T)$.

Each subject is indexed with a prognostic score, $\eta_0(\boldsymbol{Z})$, which reflects death risk in the absence of treatment; e.g., the linear predictor from a regression model. We will describe how to estimate $\eta_0(\boldsymbol{Z})$ in Sect. 2.2. Having defined $D^j(T)$ for $j = 0, 1$, conditional survival functions (given η_0) corresponding to $D^j(T)$ are given by

$$S^j(t; T|\eta_0(\boldsymbol{Z}), T) = P(D^j(T) > t|\eta_0(\boldsymbol{Z}), T) \qquad j = 0, 1.$$

The average survival function pertinent to the ATT is denoted by

$$S^j(t) = E[S^j(t; T|\eta_0(\boldsymbol{Z}), T)] \qquad j = 0, 1, \tag{1}$$

where the expectation is with respect to the joint distribution of $(\eta_0(\boldsymbol{Z}), T)$ over $\mathcal{R} \times [0, \tau]$ and $[0, \tau]$ is the range of treatment times to be considered. Note that we set τ_C such that $P(C \geq \tau_C) > 0$, with τ_C typically set to the maximum observed follow-up time. It is naturally required that $\tau < \tau_C$. We then define the ATT, with respect to the survival functions from (1) by

$$\delta(t) = S^1(t) - S^0(t), \tag{2}$$

for $t \in [0, \tau_1]$, where $\tau_1 + \tau \leq \tau_C$ for identifiability (see Schaubel and Cai 2004). It is possible that the survival functions cross at some point during the range of t under consideration. Motivated by this consideration, we propose to define the ATT through the restricted mean survival time (RMST),

$$\Delta(L) = \mu^1(L) - \mu^0(L), \tag{3}$$

where the RMSTs are given by

$$\mu^j(L) = \int_0^L S^j(t)dt \qquad j = 0, 1, \tag{4}$$

where L is a pre-specified constant. The $\mu^j(L)$ can be interpreted as the mean of the truncated residual survival time, $D^j(T) \wedge L$. In our context, one would choose $L < \tau_1$. For example, if $\tau_1 = 6$ years, many investigators would set $L = 5$ for ease of interpretation and to avoid the estimator of $\Delta(L)$ being too dependent on the behavior in the tail of the observation time distribution.

2.2 Outline of proposed method

One basic difficulty in our motivating setting is that at most one of $D_i^1(T_i)$ and $D_i^0(T_i)$ is observed. In particular, the ATTs defined in (2) and (3) involve taking a weighted average across treated patients, but (residual) survival time in the absence-of-treatment cannot be observed for a treated patient. Our solution is to match each treated subject, at the time of treatment, to a 'similar' untreated (i.e., not-yet-treated) subject.

The proposed method consists of the following steps:

(a) Estimate the prognostic score $\eta_0(\boldsymbol{Z}_i)$ based on untreated-deaths $I(D_i \leq C_i \wedge T_i)$, untreated follow-up $D_i \wedge C_i \wedge T_i$ and covariate $\boldsymbol{Z}_i$.
(b) Match each treated subject to $\leq m$ not-yet-treated subjects; e.g., $m = 5$, $m = 10$.
(c) Assign a weight to each matched set, to account for the fact that a censored data version of the T distribution is observed.

(d) Pool the matched sets, then separate into treated and untreated groups. Inverse weight the risk sets contributions of each subject; as we will show, the form of the inverse weight is different for treated and untreated subjects.
(e) Compute weighted Nelson–Aalen estimators for each treatment group, followed by the ATTs given in (2) and (3).

We now describe each of these steps in turn, making additional remarks along the way.

2.2(a) Prognostic score

The prognostic score (Prentice and Breslow 1978; Hansen 2008) is intended to represent a subject's death risk, in the absence of treatment, corresponding to the subject's adjustment covariate pattern. We assume that the pre-treatment death hazard is given by the following proportional hazards model (Cox 1972),

$$\lambda_i^D(t) = \lambda_0^D(t)\exp\{\boldsymbol{\beta}_D'\boldsymbol{Z}_i\}, \tag{5}$$

where $\lambda_0^D(t)$ is an unspecified baseline hazard and $\boldsymbol{\beta}_D$ is an unknown parameter vector. We assume that each of C_i and T_i censor D_i randomly, given $\boldsymbol{Z}_i$. This assumption on T_i requires ignorability of the treatment assignment mechanism. Specifically, we assume that $\lambda_i^D(t|\boldsymbol{Z}_i) = \lambda_i^D(t|\boldsymbol{Z}_i, C_i > t, T_i > t, U_i)$, where U_i represents any unmeasured covariate.

The parameter $\boldsymbol{\beta}_D$ can be estimated through standard partial likelihood (Cox 1975) using observed data $(X_i \wedge T_i, N_i(X_i \wedge T_i, \boldsymbol{Z}_i)$. That is, only pre-treatment follow-up is used to fit model (5). Based on (5), we would set $\eta_0(\boldsymbol{Z}_i) = \boldsymbol{\beta}_D'\boldsymbol{Z}_i$. Note that any semiparametric model could be used in place of (5). For instance, if the additive hazards model of Lin and Ying (1994) (e.g., $\lambda_0^D(t) + \boldsymbol{\beta}_D'\boldsymbol{Z}_i$), one would still set $\eta_0(\boldsymbol{Z}_i) = \boldsymbol{\beta}_D'\boldsymbol{Z}_i$. The key is that $\eta_0(\boldsymbol{Z}_i)$ does not depend on the infinite-dimensional $\lambda_0^D(t)$.

2.2(b) Matching

Consider two patients: patient k receives treatment at time T_k and patient i is alive, uncensored and not treated as of time T_k. We define the degree of similarity of these two patients through the absolute difference in prognostic scores,

$$\phi_{ik} = |\eta_0(\boldsymbol{Z}_i) - \eta_0(\boldsymbol{Z}_k)|.$$

The closer ϕ_{ik} is to 0, the more similar the potential match i is to treated patient k. A necessary condition for patient i to be selected as a match for patient k is $\phi_{ik} \le \xi$, where $\xi > 0$ is a pre-specified constant; e.g., $\xi = 0.03$. We can then define an $n \times n$ grid matrix of matching indicators, with elements

$$m_{ik} = N_k^T(\tau)I\{X_i \wedge T_i > T_k, \quad \phi_{ik} \le \xi\} \times I_{ik}$$

where I_{ik} is a 0/1 selection indicator. To clarify, set m equal to the maximum number of matches per treated subject. In cases where there are $> m$ potential matches, m eligible controls will be randomly selected; hence the need for I_{ik}. The total number of matches for T_k is then denoted by

$$m_{\bullet k} = \sum_{i=1}^{n} I(i \neq k) m_{ik},$$

with $m_{\bullet k} \leq m$. As implied, the number of matches per treated patient need not be equal across treated patients.

The algorithm entails matching-with-replacement, in that patients can be matched to multiple treated patients. Further details for the evaluation and recommendations for the choice of m are provided in Sects. 4 and 5.

2.2(c) Weight (static) assigned to matched sets at treatment time

Each matched set contains one treated patient and $\leq m$ controls, with each matched set assembled at the index patient's treatment time. Since T is subject to right censoring by C, we assign a fixed weight at the time of treatment (Horvitz and Thompson 1952). This way, the inverse-weighted distribution of T reflects that which would have been observed in the absence of censoring. Consider patient k, treated at time T_k. The fixed weight assigned to this patient (and, hence, each matched control) is given by $N_k^T(\tau) G_k(T_k)^{-1}$, where $G_i(t) = P(C_i > t | \boldsymbol{Z}_i)$. In order to estimate $G_i(t)$, C is assumed to follow a proportional hazards model,

$$\lambda_i^C(t) = \lambda_0^C(t) \exp\{\boldsymbol{\beta}_C' \boldsymbol{Z}_i\}. \tag{6}$$

Note that (6) can be estimated through standard partial likelihood, assuming that $\lambda_i^C(t|\boldsymbol{Z}_i) = \lambda_i^C(t|\boldsymbol{Z}_i, D_i > t, D_i, U_i)$, with U_i again representing any unmeasured factor (i.e., no unmeasured confounders for C).

We allow the number of matched controls to be unequal across matched sets. If one-to-one matching had been employed (as in Li et al. 2014) then the prognostic score distribution in the pooled treated and pooled untreated subjects would be equal. Without further modification, such equality is jeopardized by allowing $m_{\bullet k}$ to differ across matched sets. A solution is to rescale the controls in each matched set to represent each treated patient's "average" control. This would re-institute balance in the prognostic scores across pooled treated and untreated subjects. If matching was done at $t = 0$, then each control in the kth matched set could be given weight $1/m_{\bullet k}$. Since, in our setting, treatment assignment (and, hence, the construction of matched sets) occurs in the presence of censoring, an inverse weighted version of the scaling of controls is employed,

$$\pi_{ik} = m_{ik}\, G_i(T_k)^{-1} \times \left\{ \sum_{\ell=1}^{n} m_{\ell k} G_\ell(T_k)^{-1} \right\}^{-1}, \tag{7}$$

which can be interpreted as an inverse-weighted analog of $m_{ik}/m_{\bullet k}$.

2.2(d) Time-dependent post-matching risk-set weight

Treated subjects: The pooled treated patients and the pooled weighted controls have the same prognostic score distribution. Our intention is to estimate the survivor function for each group nonparametrically. However within each of the two pooled groups, subjects are not identically distributed. This opens the door for dependent censoring, even though (as described previously) the standard covariate-dependent censoring is assumed. Therefore, we inverse weight the post-matching data, with the weight accumulating from 1 at the index patient's treatment time (Schaubel et al. 2009). For each treated patient, the time-dependent weight is given by

$$\frac{G_k(T_k)}{G_k(T_k+t)} I(X_k > T_k + t), \tag{8}$$

such that, adapting Robins and Rotnitzky (1992), each risk set indicator is inverse weighted with respect to $P(C > T_k + t | C > T_k, \boldsymbol{Z}_k)$.

In summary, combining (7) and (8), the weight applied to each patient i, for the purposes of estimating $S^1(t)$, is given by

$$W_i^1(t) = N_i^T(\tau) I(m_{\bullet i} > 0) \frac{Y_i(T_i + t)}{G_i(T_i + t)}, \tag{9}$$

with $I(m_{\bullet i} > 0)$ indicating the exclusion of treated-but-unmatched patients.
Untreated subjects: Note that, unlike treated patients, matched controls are subject to two sources of censoring: T and C. Hence, inverse weighting with respect to both sources of censoring is required, such that the time-dependent component of the weight function for untreated patients is then given by (e.g., for untreated patient i being matched to treated patient k),

$$I(T_i \wedge X_i > T_k + t) \frac{G_i(T_k)}{G_i(T_k + t)} \exp\left\{ \int_{T_k}^{T_k+t} d\Lambda_i^T(s) \right\}, \tag{10}$$

where $\Lambda_i^T(t) = \int_{(0,t]} \lambda_i^T(u) du$. The middle element of (10) represents the conditional probability, $P(T_i > T_k + t | T_i > T_k, T_k, \boldsymbol{Z}_i)$. In order to estimate this quantity, we assume that T follows the Cox model,

$$\lambda_i^T(t) = \lambda_0^T(t) \exp\{\boldsymbol{\beta}_T' \boldsymbol{Z}_i\}, \tag{11}$$

which can be consistently estimated based on data $\{X_i \wedge T_i, \Delta_i^T, \boldsymbol{Z}_i\}$ for $i = 1, \ldots, n$, under the assumption that $\lambda_i^T(t|\boldsymbol{Z}_i) = \lambda_i^T(t|\boldsymbol{Z}_i, C_i > t, D_i > t, D_i, U_i)$; that is, we assume no unmeasured confounders for T_i.

Combining (7) and (10), the weight applied to patient i for estimating $S^0(t)$, is given by

$$W_{ik}^0(t) = \frac{N_k^T(\tau)}{G_i(T_k+t)} \, I(T_i \wedge X_i > T_k + t) \, \exp\left\{\int_{T_k}^{T_k+t} d\Lambda_i^T(s)\right\} \times \pi_{ik}. \tag{12}$$

This weight is similar to that employed in Li et al. (2014), with the notable exception of the π_{ik} component.

2.2(e) Survival function estimators

Having determined the applicable weight functions, we estimate the post-treatment cumulative hazard function through the following weighted Nelson–Aalen estimator,

$$\widehat{\Lambda}_1(t) = \sum_{i=1}^{n} \int_0^t \overline{W}_1(s)^{-1} \widehat{W}_i^1(s) dN_i(T_i + s), \tag{13}$$

where $\overline{W}_1(s) = \sum_{\ell=1}^{n} \widehat{W}_\ell^1(s)$. Having computed (13), $S_1(t)$ is then estimated by $\widehat{S}_1(t) = \exp\{-\widehat{\Lambda}_1(t)\}$. The absence-of-treatment survival function estimator is given by $\widehat{S}_0(t) = \exp\{-\widehat{\Lambda}_0(t)\}$, where

$$\widehat{\Lambda}_0(t) = \sum_{k=1}^{n} \sum_{i=1}^{n} \int_0^t \overline{W}_1(s)^{-1} \widehat{W}_{ik}^0(s) dN_i(T_k + s) N_k^T(\tau), \tag{14}$$

where $\overline{W}_1(s) = \sum_{k=1}^{n} \sum_{\ell=1}^{n} \widehat{W}_{\ell k}^0(s) N_k^T(\tau)$.

The treatment effect estimator is then given by

$$\widehat{\delta}(t) = \widehat{S}_1(t) - \widehat{S}_0(t)$$

with respect to the survival function, and by

$$\widehat{\Delta}(L) = \widehat{\mu}_1(L) - \widehat{\mu}_0(L)$$
$$\widehat{\mu}_j(L) = \int_0^L \widehat{S}_j(t) dt,$$

with respect to RMST.

A few additional comments on our assumptions are in order. First, it should be noted that we assume $P(D^0(T) > t|\eta_0(\mathbf{Z}), T) = P(D^0(T) > t|\mathbf{Z}, T)$. However, it is not required that $P(D^1(T) > t|\eta_0(\mathbf{Z}), T) = P(D^1(T) > t|\mathbf{Z}, T)$. Second, since non-linear models are biased in the presence of unmeasured predictors, we require the correctness of models. With respect to positivity, we assume that $P(T > t|\eta_0(\mathbf{Z})) > 0$ for all $t \in [0, \tau]$, such that matches are available for any treatment that could occur within the time subinterval of interest.

3 Variance estimator

We propose a closed form variance formula for our point estimators. The aim is to derive a convenient formula which captures the most important aspects of the variability in the proposed estimators. Hence, for the purposes of variance estimation, we treat the weight function and matching mechanism as fixed. Note that the practice of estimating standard errors with the inverse weight as fixed has a long history in causal inference, dating back to at least Robins et al. (2000). Explicitly accommodating the matching process in the variance derivation is known to be quite complex; our setting is no exception. This variance estimator will be evaluated numerically in Sect. 4.

With respect to $\widehat{S}^1(t)$, it is well-known that $n^{1/2}\{\widehat{\Lambda}_1(t)-\Lambda_1(t)\} = n^{-1/2}\sum_{i=1}^n \Psi_i^1(t)$ asymptotically (Fleming and Harrington 1991), where

$$\Psi_i^1(t) = \int_0^t \overline{w}_1(u)^{-1} W_i^1(u) dM_i^1(u), \tag{15}$$

with $\overline{w}_1(u) = E[W_i^1(u)]$ and $dM_i^1(u) = dN_i^1(u) - I(X_i \geq T_i + u)d\Lambda_1(u)$. Under mild regularity conditions, the $\{\Psi_i^1(t), \cdots, \Psi_n^1(t)\}$ are independent and identically distributed mean 0 variates. Applying the Multivariate Central Limit Theorem, $n^{1/2}\{\widehat{\Lambda}_1(t) - \Lambda_1(t)\}$ converges asymptotically to a mean-zero Normal distribution with variance $E[\Psi_i^1(t)^2]$. Then, applying the Functional Delta Method (van der Vaart 1998), $n^{1/2}\{\widehat{S}^1(t) - S^1(t)\}$ is asymptotically mean-zero Normal with variance estimator,

$$\widehat{\sigma}_1^2(t) = n^{-1}\sum_{i=1}^n \{\widehat{S}_1(t)\widehat{\Psi}_i^1(t)\}^2,$$

where

$$\widehat{\Psi}_{i1}(t) = \int_0^t \overline{W}_1(u)^{-1}\widehat{W}_i^1(u) d\widehat{M}_i^1(u),$$

with $d\widehat{M}_i^1(u) = dN_i^1(u) - I(X_i \geq T_i + u)d\widehat{\Lambda}_1(u)$.

Now, with respect to the asymptotic properties of $\widehat{S}^0(t)$, one difficulty is that $\widehat{\Lambda}_0(t)$ is not a summation of independent variates. As such, direct application of the classical limit theory for sums of independent variables is not available. However, after applying Slutsky's Theorem, the proposed estimator can be treated, in large samples, as a U-statistic (Hoeffding 1948), which can in turn be approximated within a sufficient degree of accuracy by a sum of independent and identically distributed variates. From this perspective, the classical limit theory for sums of does carry over to $\widehat{\Lambda}_0(t)$ to yields the relevant asymptotic properties. Specifically, applying Theorem 5.3.3 of Serfling (1980) implies that $n^{1/2}\{\widehat{\Lambda}_0(t) - \Lambda_0(t)\}$ converges asymptotically to a mean zero

Normal variate with variance $E[\Psi_i^0(t)^2]$, where

$$\Psi_i^0(t) = \sum_{k=1}^{n} N_k^T(\tau) \int_0^t \overline{w}_0(u)^{-1} W_{ik}^0(u) dM_{ik}^0(u), \tag{16}$$

with $\overline{w}_0(u) = E[W_{ik}^0(u)]$ and $dM_{ik}^0(u) = dN_{ik}^0(u) - I(X_i \geq T_k + u) d\Lambda_0(u)$. Note that the potential repetition of a subject across different matched sets is accommodated by aggregating across k in (16). Applying the Functional Delta Method (van der Vaart 1998), $n^{1/2}\{\widehat{S}_0(t) - S_0(t)\}$ is also asymptotically mean-zero Normal with variance estimator

$$\widehat{\sigma}_0^2(t) = n^{-1} \sum_{i=1}^{n} \{\widehat{S}^0(t) \widehat{\Psi}_i^0(t)\}^2,$$

where

$$\widehat{\Psi}_i^0(t) = \sum_{k=1}^{n} N_k^T(\tau_T) \int_0^t \overline{W}_0(u)^{-1} \widehat{W}_{ik}^0(u) d\widehat{M}_{ik}^0(u).$$

Combining (15) and (16), it follows that $n^{1/2}\{\widehat{\delta}(t) - \delta(t)\}$ is also asymptotically mean-zero Normal with variance estimator

$$\widehat{\sigma}_\delta^2(t) = n^{-1} \sum_{i=1}^{n} \{\widehat{S}^1(t) \widehat{\Psi}_i^1(t) - \widehat{S}^0(t) \widehat{\Psi}_i^0(t)\}^2.$$

Integrating the expansions (15) and (16), it follows that $n^{1/2}\{\widehat{\Delta}(L) - \Delta(L)\}$, is asymptotically Normal with variance estimator

$$\widehat{\sigma}_\Delta^2(L) = n^{-1} \sum_{i=1}^{n} \{\widehat{\Gamma}^1(L) - \widehat{\Gamma}^0(L)\}^2,$$

where

$$\widehat{\Gamma}^j(L) = \int_0^L \{\widehat{\Psi}_i^j(t) \widehat{S}^j(t)\} dt \qquad j = 0, 1.$$

4 Simulation study

Finite-sample properties of the proposed estimators are evaluated through a series of simulation studies. First, the covariate vector for each subject was generated as $\boldsymbol{Z}_i = (Z_{i1}, Z_{i2}, Z_{i3})'$, where Z_{i1} is a Bernoulli(0.5) variate, and $(Z_{i2}, Z_{i3})'$ arises from a bivariate Normal distribution with mean $(0.5, -0.5)'$, variances both 1 and covariance 0.25. Treatment times T_i were generated from an exponential distribution

with rate $\lambda_i^T(t) = \lambda_0^T \exp\{\boldsymbol{\beta}_T' \boldsymbol{Z}_i\}$, with $\boldsymbol{\beta}_T = [\log(0.75), \log(0.7), \log(0.75)]'$. The counterfactual D^0 was generated from an exponential distribution with rate $\lambda_i^0(t) = \lambda_0^0 \exp\{\boldsymbol{\beta}_0' \boldsymbol{Z}_i\}$, where $\boldsymbol{\beta}_0 = [\log(1.3), \log(1.25), \log(1.25)]'$. Censoring times were generated from a piecewise exponential model based on the following subintervals: [0, 1], (1, 2], (2, 4], (4, 8], (8, 10]. For the kth interval, the hazard function is given by $\lambda_k^C \exp\{\boldsymbol{\beta}_C' \boldsymbol{Z}_i\}$, for $k = 1, \ldots, 5$ and $\boldsymbol{\beta}_C = [\log(0.8), \log(1.25), \log(1.3)]'$.

We varied the sample size to be $n = 500$ and $n = 2000$. We have every subject's potential time to treatment, time to death (in the presence and in the absence of treatment) and time to censoring. We set $\tau = 3$, $L = 5$ and $\xi = \log(1.05)$. We varied the maximum number of matches for each treated subject to be $m = 1$ (1:1 matching) and $m = 5$ (1:5 matching). In cases where there were > 5 matched subjects for a treated subject, we randomly choose 5 matched subjects.

We evaluated many data configurations following the above scheme. Results presented in Tables 1 and 2 correspond to settings where the post-treatment residual survival time $D^1(T)$ followed a piecewise exponential distribution with rate $\lambda_i^1(t) = \lambda_0^0 \exp\{\boldsymbol{\beta}_0' \boldsymbol{Z}_i + \alpha(t)\}$, where $\alpha(t) = -0.1\{1 + 0.2I(t - T < 0.5)\}$; this results in a moderate treatment effect.

We conducted 500 replicates for each scenario. Tables 1 and 2 report 6 measures: bias, empirical standard deviation (ESD), asymptotic standard error (ASE), empirical coverage probability (ECP), square root of mean square error (Root MSE), and relative efficiency (e.g., ratio of MSE of 1:1 match divided by MSE of 1:5 match). Across all time points, the bias is relatively small, the ASE is quite close to the ESD and, correspondingly, ECPs are quite consistent with the nominal level of 0.95. Slight under-coverage is present, possibly due to our not having accounted for the randomness in the inverse weights and/or matching mechanism. Compared with 1:1 matching, 1:5 matching yields substantially greater efficiency.

In additional simulation runs, the proposed methods performed well when the prognostic score was replaced by a propensity score (Lu 2005) as a basis for matching, although prognostic scoring appeared to have slightly less residual bias and to be moderately more efficient (Table 3; the simulation set-up was the same as Table 2). In the Supplementary Materials, we evaluate the robustness of the proposed methods to model mis-specification. It appears that the methods are fairly robust to mis-specifying the C or T hazard models, or the prognostic score model.

5 Analysis of liver transplant data

We applied the proposed methods to estimate the effect of deceased-donor kidney transplantation (treatment) on survival compared to control (without receiving kidney transplantation) among patients who received a liver transplant. Data were obtained from the Scientific Registry of Transplant Recipients (SRTR). The SRTR data system includes data on all donors, wait-listed candidates, and transplant recipients in the U.S., as submitted by the members of the Organ Procurement and Transplantation Network (OPTN), and has been described elsewhere. The Health Resources and

Table 1 Simulation results: moderate treatment effect; $n = 500$

1:m	Measure	t	TRUE	BIAS	ESD	ASE	ECP	Root MSE	Efficiency
1:1	$\widehat{S}_1(t)$	1	0.796	−0.003	0.033	0.033	0.94	0.045	1
		3	0.643	−0.007	0.040	0.039	0.96	0.054	1
		5	0.521	−0.007	0.042	0.041	0.95	0.057	1
	$\widehat{S}_0(t)$	1	0.818	−0.004	0.036	0.036	0.95	0.046	1
		3	0.558	−0.008	0.061	0.059	0.93	0.067	1
		5	0.386	−0.011	0.076	0.071	0.91	0.079	1
	$\widehat{\delta}(t)$	1	−0.020	0.001	0.049	0.049	0.94	0.066	1
		3	0.086	0.001	0.073	0.071	0.94	0.097	1
		5	0.134	0.004	0.086	0.082	0.93	0.114	1
	$\widehat{\mu}_1(t)$	5	3.464	−0.028	0.158	0.157	0.95	0.214	1
	$\widehat{\mu}_0(t)$	5	3.172	−0.036	0.219	0.210	0.94	0.293	1
	$\widehat{\Delta}(t)$	5	0.292	0.007	0.269	0.262	0.95	0.360	1
1:5	$\widehat{S}_1(t)$	1	0.796	−0.002	0.033	0.032	0.94	0.044	1.027
		3	0.643	−0.004	0.039	0.039	0.95	0.054	1.008
		5	0.521	−0.004	0.042	0.041	0.92	0.057	1.017
	$\widehat{S}_0(t)$	1	0.818	−0.003	0.022	0.021	0.94	0.038	2.928
		3	0.558	−0.006	0.041	0.041	0.95	0.056	2.031
		5	0.386	−0.006	0.055	0.052	0.92	0.064	1.901
	$\widehat{\delta}(t)$	1	−0.020	0.001	0.039	0.039	0.96	0.053	1.573
		3	0.086	0.002	0.055	0.056	0.96	0.077	1.594
		5	0.134	0.001	0.067	0.067	0.95	0.090	1 .615
	$\widehat{\mu}_1(t)$	5	3.464	−0.017	0.156	0.156	0.95	0.212	1.025
	$\widehat{\mu}_0(t)$	5	3.172	−0.023	0.155	0.149	0.94	0.207	1.325
	$\widehat{\Delta}(t)$	5	0.292	0.006	0.210	0.215	0.95	0.291	1.537

Services Administration (HRSA), U.S. Department of Health and Human Services provides oversight to the activities of the OPTN and SRTR contractors.

As depicted in Fig. 2, time 0 for a patient was the time of liver transplantation (LT), and the treatment being evaluated was subsequent receipt of a kidney transplant; i.e., so-called kidney-after-liver transplantation (KALT). Specifically, the study population included the $n = 57,177$ patients aged ≥ 18 years who received a deceased-donor LT between 03/01/2002 and 12/31/2014. Patients began follow-up at the date of LT and were followed until the earliest of death, loss to follow-up, or the end of the observation period (12/31/2014).

A total of 763 patients were observed to receive KALT. Among patients who received kidney transplantation, there were 74 post-transplant deaths and 689 patients censored. For each of the treatment-free hazard, transplantation hazard, and censoring hazard models, adjustment covariates included age, sex, race, diagnosis, initial dialysis modality, and calendar year of LT. For the prognostic score matching, we set

Table 2 Simulation results: moderate treatment effect; $n = 2000$

1:m	Measure	t	TRUE	BIAS	ESD	ASE	ECP	Root MSE	Efficiency
1:1	$\widehat{S}_1(t)$	1	0.798	−0.003	0.016	0.016	0.97	0.022	1
		3	0.643	−0.006	0.019	0.020	0.95	0.027	1
		5	0.521	−0.005	0.020	0.021	0.94	0.028	1
	$\widehat{S}_0(t)$	1	0.818	−0.002	0.018	0.018	0.96	0.023	1
		3	0.558	−0.006	0.030	0.031	0.97	0.034	1
		5	0.386	−0.006	0.041	0.039	0.93	0.041	1
	$\widehat{\delta}(t)$	1	−0.020	−0.001	0.023	0.024	0.96	0.033	1
		3	0.086	0.001	0.036	0.036	0.96	0.049	1
		5	0.134	0.001	0.046	0.044	0.94	0.061	1
	$\widehat{\mu}_1(t)$	5	3.464	−0.021	0.075	0.078	0.96	0.106	1
	$\widehat{\mu}_0(t)$	5	3.172	−0.023	0.111	0.109	0.95	0.150	1
	$\widehat{\Delta}(t)$	5	0.292	0.002	0.132	0.134	0.96	0.182	1
1:5	$\widehat{S}_1(t)$	1	0.798	−0.003	0.016	0.016	0.97	0.022	1.010
		3	0.643	−0.006	0.019	0.020	0.95	0.027	1.023
		5	0.521	−0.005	0.021	0.021	0.95	0.028	0.985
	$\widehat{S}_0(t)$	1	0.818	−0.002	0.010	0.011	0.95	0.019	2.944
		3	0.558	−0.006	0.021	0.021	0.94	0.028	2.001
		5	0.386	−0.006	0.029	0.028	0.92	0.033	1.914
	$\widehat{\delta}(t)$	1	−0.020	−0.000	0.018	0.019	0.95	0.026	1.598
		3	0.086	0.001	0.028	0.029	0.97	0.038	1.650
		5	0.134	0.001	0.034	0.035	0.95	0.047	1.669
	$\widehat{\mu}_1(t)$	5	3.464	−0.021	0.073	0.078	0.95	0.105	1.018
	$\widehat{\mu}_0(t)$	5	3.172	−0.023	0.077	0.077	0.93	0.107	1.961
	$\widehat{\Delta}(t)$	5	0.292	0.002	0.103	0.109	0.97	0.145	1.578

$\xi = \log(1.05)$. We evaluated KALT occurring within $\tau = 5$ years of LT. We chose the maximum number of matches for each KALT patient to be $m = 5$.

Figure 4 shows the average survival curve, $\widehat{S}^1(t)$, from the time of KALT among patients, as well as the absence-of-KALT survival curve, $\widehat{S}^0(t)$. The estimated $\widehat{S}^1(t)$ was 0.99, 0.97, and 0.94, at $t = 1$, $t = 3$ and $t = 5$ years, respectively. In contrast, the estimated $\widehat{S}^0(t)$ was 0.95, 0.88 and 0.82 at the same time points. The ATT for survival was estimated to be $\widehat{\delta}(1) = 0.04$, $\widehat{\delta}(3) = 0.08$ and $\widehat{\delta}(5) = 0.12$, respectively.

We evaluated the ATT estimator with respect to $\Delta(5)$ in Table 4 for the following maximum number of matches: $m = 1$, $m = 5$, and $m = 10$. Comparing with 1:1 matching, setting $m = 5$ and $m = 10$ yielded substantial improvements in efficiency. Important gains in efficiency were observed going from $m = 5$ to $m = 10$, although returns did diminish for $m > 10$ (data not tabulated). In any case, $\widehat{\Delta}(5) \approx 0.26$, indicting that the gain in restricted (i.e., capped at 5 years) residual survival time post-KALT is approximately 3 months, averaged across KALT recipients.

Table 3 Comparison of prognostic score matching and propensity score matching

Method	Measure	t	TRUE	BIAS	ESD	ASE	ECP	Root MSE	Efficiency
Prognostic	$\widehat{S}_1(t)$	1	0.798	−0.003	0.016	0.016	0.97	0.022	1
		3	0.643	−0.006	0.019	0.020	0.95	0.027	1
		5	0.521	−0.005	0.021	0.021	0.95	0.028	1
	$\widehat{S}_0(t)$	1	0.818	−0.002	0.010	0.011	0.95	0.019	1
		3	0.558	−0.006	0.021	0.021	0.94	0.028	1
		5	0.386	−0.006	0.029	0.028	0.92	0.033	1
	$\widehat{\delta}(t)$	1	−0.020	−0.000	0.018	0.019	0.95	0.026	1
		3	0.086	0.001	0.028	0.029	0.97	0.038	1
		5	0.134	0.001	0.034	0.035	0.95	0.047	1
	$\widehat{\mu}_1(t)$	5	3.464	−0.021	0.073	0.078	0.95	0.105	1
	$\widehat{\mu}_0(t)$	5	3.172	−0.023	0.077	0.077	0.93	0.107	1
	$\widehat{\Delta}(t)$	5	0.292	0.002	0.103	0.109	0.97	0.145	1
Propensity	$\widehat{S}_1(t)$	1	0.798	−0.002	0.015	0.016	0.96	0.022	0.987
		3	0.643	−0.004	0.020	0.020	0.95	0.026	0.995
		5	0.521	−0.005	0.020	0.021	0.96	0.028	0.963
	$\widehat{S}_0(t)$	1	0.818	−0.002	0.012	0.012	0.95	0.020	0.746
		3	0.558	−0.006	0.023	0.023	0.94	0.029	0.855
		5	0.386	−0.007	0.031	0.030	0.4	0.034	0.886
	$\widehat{\delta}(t)$	1	−0.020	0.000	0.019	0.020	0.96	0.027	0.907
		3	0.086	0.002	0.029	0.030	0.97	0.040	0.914
		5	0.134	0.002	0.036	0.037	0.95	0.049	0.916
	$\widehat{\mu}_1(t)$	5	3.464	−0.015	0.074	0.078	0.96	0.104	0.976
	$\widehat{\mu}_0(t)$	5	3.172	−0.021	0.082	0.083	0.96	0.114	0.889
	$\widehat{\Delta}(t)$	5	0.292	0.007	0.105	0.113	0.96	0.149	0.937

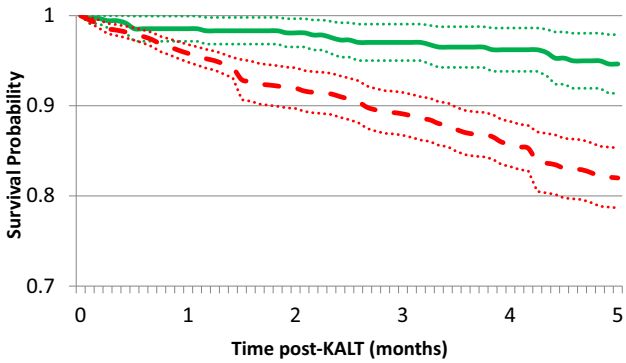

Fig. 4 Analysis of liver transplant data: estimated survival curves and 95% CIs for 5-year post-KALT (solid line) and absence-of-KALT (dashed line)

Table 4 Analysis of SRTR data: results based on $L = 5$ years

1:m	$\widehat{\mu}_1(5)$			$\widehat{\mu}_0(5)$			$\widehat{\Delta}(5)$		
	Estimate	SE	Efficiency	Estimate	SE	Efficiency	Estimate	SE	Efficiency
1:1	4.790	0.039	1	4.532	0.080	1	0.258	0.089	1
1:5	4.790	0.039	1	4.516	0.032	6.086	0.274	0.050	3.097
1:10	4.790	0.039	1	4.562	0.020	16.484	0.263	0.043	4.173
1:25	4.790	0.039	1	4.593	0.019	18.125	0.201	0.042	4.199

6 Discussion

We aim to compare post-treatment survival with the average survival function that would have been observed in the absence of treatment. In our motivating example, treatment is time-dependent in the sense that subjects typically begin untreated, with some going on to receive treatment at some time point. In addition, we are primarily considering observational studies in which treatment is not assigned at random and, therefore, may depend on various baseline patient characteristics. We have developed a semi-parametric matching method to estimate the causal effect of treatment among the treated with respect to survival probability and restricted mean survival time. Under the proposed methods, each patient is matched to as-yet-untreated patients with similar prognostic scores. The experience of treated and matched patients is weighted using a variant of Inverse Probability of Censoring Weighting, after which pre- and post-treatment weighted Nelson–Aalen survival curves are estimated and contrasted. A closed-form variance estimator is proposed and shown to perform well in finite samples. An application to kidney-after-liver transplantation reveals a 3-month gain in RMST (based on a 5-year time horizon) among KALT patients.

Our estimand is causal average effect of treatment on the treated. A potential disadvantage of the ATT is the fact that it is not invariant to the treatment time distribution. However, the degree to which this is a limitation is setting dependent. It could be argued that the ATT is ideal for public health/policy purposes, particularly in the context of organ transplantation. For example, if comparing two regions within a country, a region that selects only the healthiest patients for transplant (i.e., those who would have favorable prognosis, whether treated or not) could have fewer patient-years added due to transplantation relative to a region that more appropriately transplanted the patients truly in need of a transplant. In any case, the ATT is a useful alternative to the average causal effect (ACE), and may be more relevant and preferred in various settings. Unlike the average causal effect (ACE), the ATT on the treated is particularly well-suited to the study of organ transplantation since it would be undesirable and in fact impossible to transplant all patients.

The matching we propose is based on the prognostic score (Hansen 2008) which can be considered an outcome risk analog of the propensity score. We use Cox regression to compute the prognostic score, throughout this report, although any semiparametric metric model could be employed; e.g., the additive hazards model of Lin and Ying (1994). Note that a propensity score could also serve as the basis for matching, such

as the time-dependent Cox regression-based propensity score used by Lu (2005). We chose the prognostic score based in part on increased efficiency. The use of a score (i.e., linear predictor) from a semiparametric model affords some flexibility with respect to model mis-specification. One might choose between the propensity score and prognostic score (as a basis for matching) based on which model discriminates (outcome risk, treatment assignment rate) more accurately.

We demonstrate great efficiency gains by using multiple (for example, and $m = 5$) matches per treated patient, compared to 1:1 matching. Efficiency gains appear to tail off after $m = 5$, at least under the settings we examined. If treatment was assigned at time 0, or if censoring did not depend on the covariate vector, IPCW would not be required in the rescaling of the matched controls. Since censoring is allowed to depend on the covariate under our assumed data structure, the weight function is more complicated and takes much more time to compute.

Our real-data application indicates a modest survival benefit for KALT. Such results add support to the more judicious use of of simultaneous kidney–liver transplantation (SKLT). However, it should be noted that we did not compare SKLT versus KALT. Our (albeit closely related) comparison was between KALT and prognostically similar patients who did not receive KALT. A formal comparison between SLKT and KALT would better address the hypothesis that the latter could adequately substitute for the former as a treatment option for patients wait-listed for liver transplantation who also have renal dysfunction.

A potential issue in applying the proposed method to the organ failure setting is the lack of independence between T_i and T_ℓ for $i \neq \ell$. In particular, only one patient will be transplanted when a given donor organ is offered, which introduces a lack of independence in the treatment times across patients. Vock et al. (2013) is among the very few works that accommodate this issue explicitly. However, this issue would be much less pervasive in our KALT example relative to other organ failure settings. Specifically, of the $> 150{,}000$ kidney transplants which occurred during our observation period, < 1000 were KALT. Therefore, the vast majority of the 'competition' for donor kidneys would have been outside our study population.

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References

Cox DR (1972) Regression models and life tables (with Discussion). J R Stat Soc Ser B 34:187–200
Cox DR, (1975) Partial likelihood. Biometrika 62:269–276
Cox DR, Oakes D (1984) The analysis of survival data. Chapman and Hall, New York
Fleming TR, Harrington DP (1991) Counting processes and survival analysis. Wiley, Chichester
Hansen BB (2008) The prognostic analogue of the propensity score. Biometrics 95(2):481–488
Hernán MA, Brumback B, Robins JM (2000) Marginal structural models to estimate the causal effect of zidovudine on the survival of HIV-positive men. Epidemiology 11:561–570

Hernán MA, Brumback B, Robins JM (2001) Marginal structural models to estimate the joint causal effect of nonrandomized treatments. J Am Stat Assoc 96:440–448
Hernán MA, Cole SR, Margolick J, Cohen M, Robins JM (2005) Structural accelerated failure time models for survival analysis in studies with time-varying treatments. Pharmacoepidemiol Drug Saf 14:477–491
Hoeffding W (1948) A class of statistics with asymptotically normal distributions. Ann Stat 19:293–325
Horvitz DG, Thompson DJ (1952) A generalization of sampling without replacement from a finite universe. J Am Stat Assoc 47:663–685
Kalbfleisch JD, Prentice RL (2002) The statistical analysis of failure time data, 2nd edn. Wiley, Hoboken
Kaplan EL, Meier P (1958) Nonparametric estimation from incomplete observations. J Am Stat Assoc 282:457–481
Li Y, Schaubel DE, He K (2014) Matching methods for obtaining survival functions to estimate the effect of a time-dependent treatment. Stat Biosci 6:105–126
Lin DY, Ying Z (1994) Semiparametric analysis of the additive risk model. Biometrika 81:61–71
Lok J, Gill R, van der Vaart A, Robins J (2004) Estiamting the causal effect of a time-varying treatment on time-to-event using structural nested failure time models. Stat Neerl 58:271–295
Lu B (2005) Propensity score matching with time-dependent covariates. Biometrics 61:721–728
Petersen ML, Deeks SG, Martin JN, van der Laan MJ (2007) History-adjusted marginal structural models for estimating time-varying effect modification. Am J Epidemiol 166:185–193
Prentice RL, Breslow NE (1978) Retrospective studies and failure time models. Biometrika 65(1):153–158
Robins JM, Hernán MA (2008) Estimation of the causal effects of time-varying exposures. In: Fitzmaurice G, Davidian M, Verbeke G, Molenberghs G (eds) Advances in longitudinal data analysis. Chapman and Hall, New York
Robins JM, Rotnitzky A (1992) Recovery of information and adjustment for dependent censoring using surrogate markers. In: Jewell N, Dietz K, Farewell V (eds) AIDS epidemiology—methodological issues. Birkhauser, Boston, pp 297–331
Robins JM, Hernán MA, Brumback B (2000) Marginal structural models and causal inference in epidemiology. Epidemiology 11:550–560
Schaubel DE, Cai J (2004) Regression methods for gap time hazard functions of sequentially ordered multivariate failure time data. Biometrika 91(2):291–303
Schaubel DE, Wolfe RA, Sima CS, Merion RM (2009) Estimating the effect of a time-dependent treatment by levels of an internal time-dependent covariate. J Am Stat Assoc 104:49–59
Serfling RJ (1980) Approximation theorems of mathematical statistics. Wiley, New York
Sharma P, Shu X, Schaubel DE, Sung RS, Magee JC (2016) Propensity score-based survival benefit of simultaneous liver-kidney transplant over liver transplant alone for recipients with pretransplant renal dysfunction. Liver Transplant 22:71–79
Struthers CA, Kalbfleisch JD (1986) Misspecified proportional hazards mdoels. Biometrika 73(2):363–369
van der Vaart AW (1998) Asymptotic statistics. Cambridge University Press, Cambridge
Vock DM, Tsiatis AA, Davidian M, Laber EB, Tsuang WM, Finlen-Copeland CA, Palmer SM (2013) Assessing the causal effect of organ transplantation on the distribution of residual lifetime. Biometrics 69(4):820–829
Wei G, Schaubel DE (2008) Estimating cumulative treatment effects in the presence of nonproportional hazards. Biometrics 64:724–732

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